

Variable Fidelity CIM Modeling

July 6, 2025

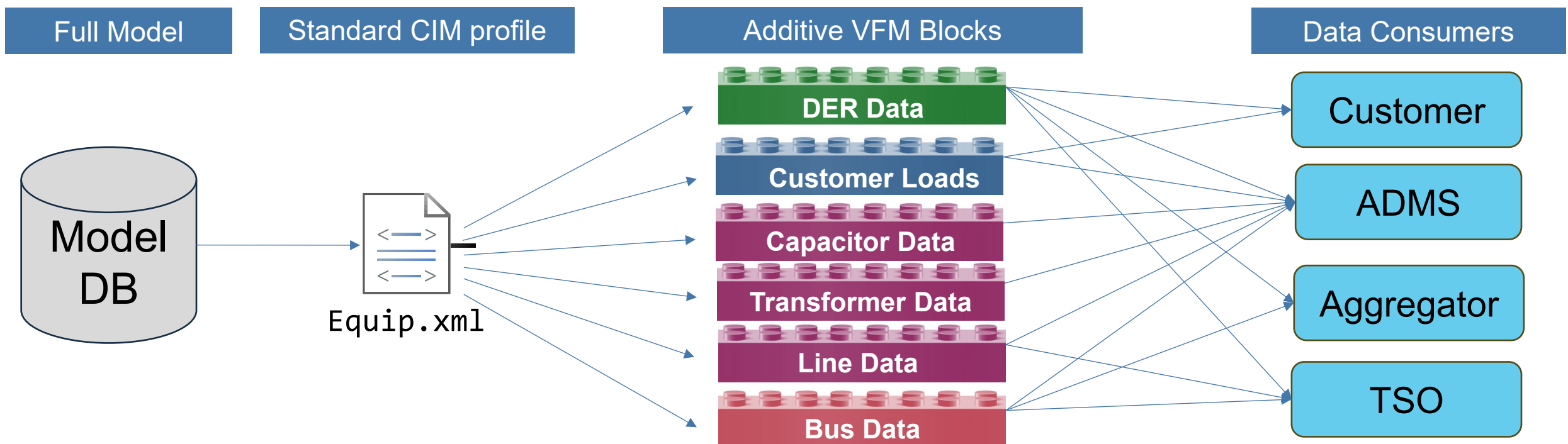
Dr. Alex A. Anderson

Dr. Rabayet Sadnan



What is Variable Fidelity Modeling?

- Existing IEC 61970 CIM profiles work well for restricting data at high-level (e.g., generator owner can receive topology, but no market bid data)
- Existing profiles *are not granular enough* to provide entity-specific model exports
- Variable Fidelity Modeling (VFM) applies a combination of semantic standards to develop attribute-based access control to zero-trust sharing of distribution grid data



Data Exchange Use Cases

From	To	Use Case	Data Exchange
DSO	Customers	DER participation in Distribution Grid Service	DR signal. Price signal, registry
Customers	DSO		
DSO	Customers	New Equipment Installation	DR signal. Price signal, registry
Customers	DSO		
DSO	DER owners	DER participation in DSO Service	injection, pf, DER registration
DSO	DER owners	Registration	injection, pf, DER registration
DSO	DER owners	Interconnection Request	injection, pf, DER registration
DSO	DER owners	Emergency Shut off	injection, pf, DER registration
DSO	DER owners	Direct Dispatch (e.g. DERMS)	injection, pf, DER registration
DSO	DERA	Distribution Impacts of DERA participation in Bulk Grid Service	capacity, price, location
DSO	DERA	DERA participation in Distribution Grid Service	capacity, price, location
DSO	DERA	Operating Envelope	capacity, price, location
DSO	DERA	Registration	capacity, price, location, registration
DSO	DERA	Emergency Shut off	control signal
TSO	DSO	DER Output (Aggregated) FERC order 901	capacity, price, location, registration
TSO	DSO	DER capacity (Aggregated) FERC order 901	capacity, price, location, registration
DSO	ADMS/OPF/ APPS		

- Exploring Multi-Entity Interactions:
 - Navigating complex connections between diverse stakeholders within systems.
 - Facilitating seamless services and cooperation among varied entities.
- Navigating Data Exchange Dynamics:
 - Understanding the flow and coordination of critical information.
 - Enhancing communication pathways for effective data transmission
- Tailored Data Format and Interoperability:
 - Adopting standardized data formats to ensure compatibility across diverse systems and platforms.
 - Facilitating seamless data integration by ensuring interoperability through agreed-upon protocols and interfaces.
- Selective Information Sharing:
 - Exchange only relevant data tailored to specific use cases.
 - Variable Fidelity (VF) Block – Extracts a subset of profile information from complex data for easier use.

Six Steps for Variable Fidelity Modeling

EAP

- Download appropriate golden-copy CIM version (16 / 17 / 18) from UCAiug website
- Add use-case specific extensions in Enterprise Architect

OWL

- Define VFM Blocks as .owl artifacts generated using CIMTool
- Combine blocks to form a custom profile using GridAPPS-D VFM Editor GUI

Schema

- Save custom exchange profile into CIMTool Workspace, which auto-generates schemas
- Save Python schemas from CIMTool into working directory

Python

- Set up master database with complete network model (XML files can also be opened directly)
- Open feeder model with CIMantic Studio desktop app or CIMantic Graphs API with custom python schema

SHACL

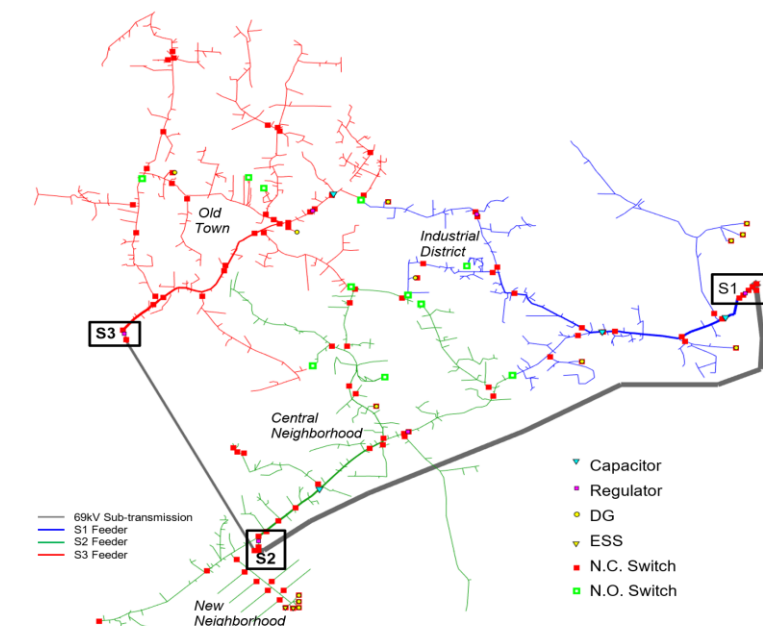
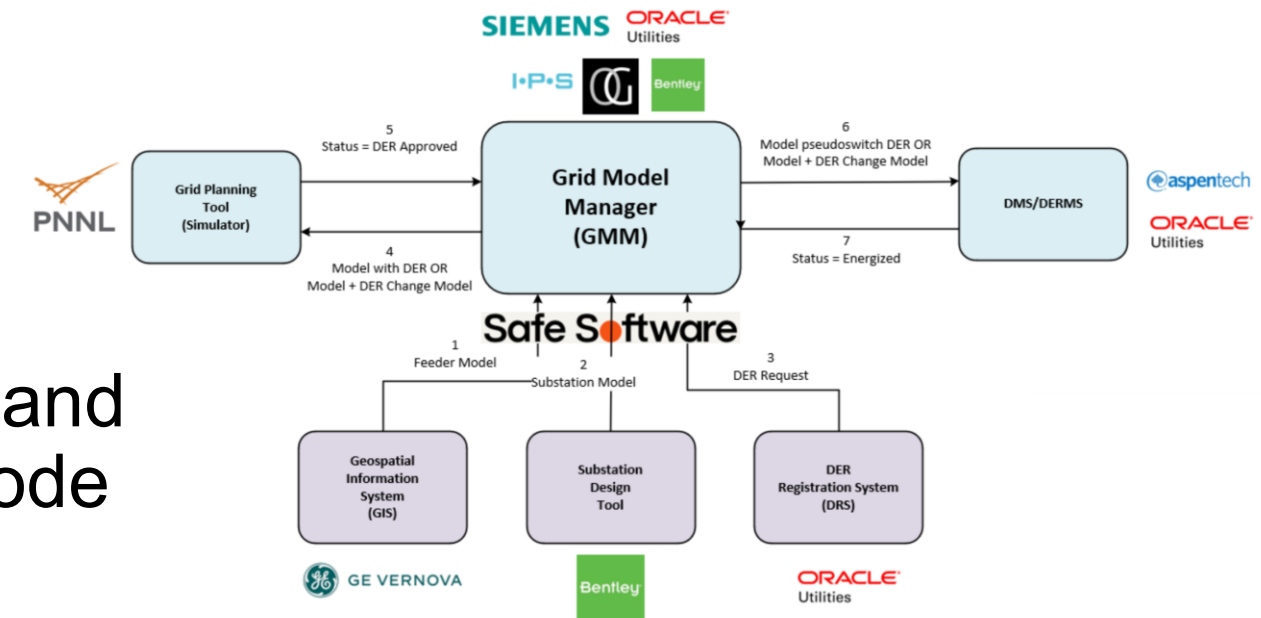
- Define custom value-based data export restrictions (e.g. only export buses connected to a DER)
- Save feeder model using CIMantic Studio or CIM-Graph API with SHACL export rules

XML

- Place exported XML into entity-specific data zone with lock-key mechanism or named database graph for retrieval by entity

Variable Fidelity Models for EPRI GMDM Interop II

- Variable Fidelity Modeling approach to be used for upcoming EPRI Grid Model Data Management interop demo
- Seven vendors to demonstrate exchange and ingest of CIM 18 version of PNNL 9500 Node Test Feeder for DER Registry use case
- New VFM blocks and exchange profiles have been defined for interop and test model
 - Unbalanced Connectivity
 - Unbalanced Electrical
 - Geospatial Location
 - Asset Catalogue
 - Market Node



Modeling Glossary



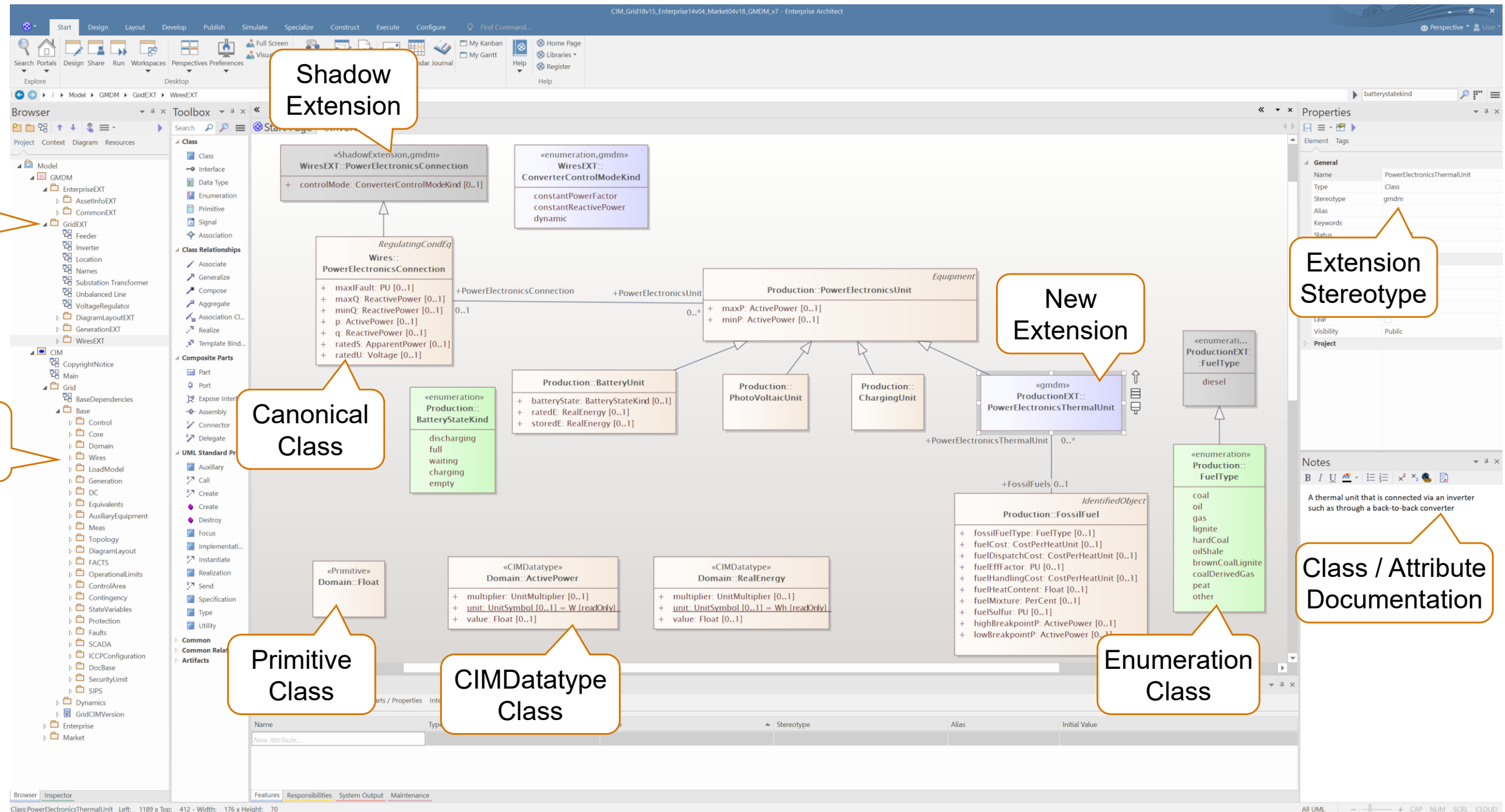
Need for New Modeling Vocabulary

- CIM community has been overusing the word “profile” leading to confusion even within internal UCAiug working groups
- Worked closely with AspenTech and CIMple Data to develop new glossary of variable fidelity modeling concepts that could be used across efforts
- Concepts align with tooling capabilities and VFM workflows developed
- Basic Concepts:
 - **Canonical Class:** Modeling concept that has been reviewed by UCAiug working groups and approved by consensus of one or more committees
 - **Extension Class:** Modeling concept that is used for custom data exchange
 - **Informative Class:** Modeling concept that is included within golden copy UML but has not been reviewed yet or has been marked to be deleted in next CIM version.

CIM Stereotypes

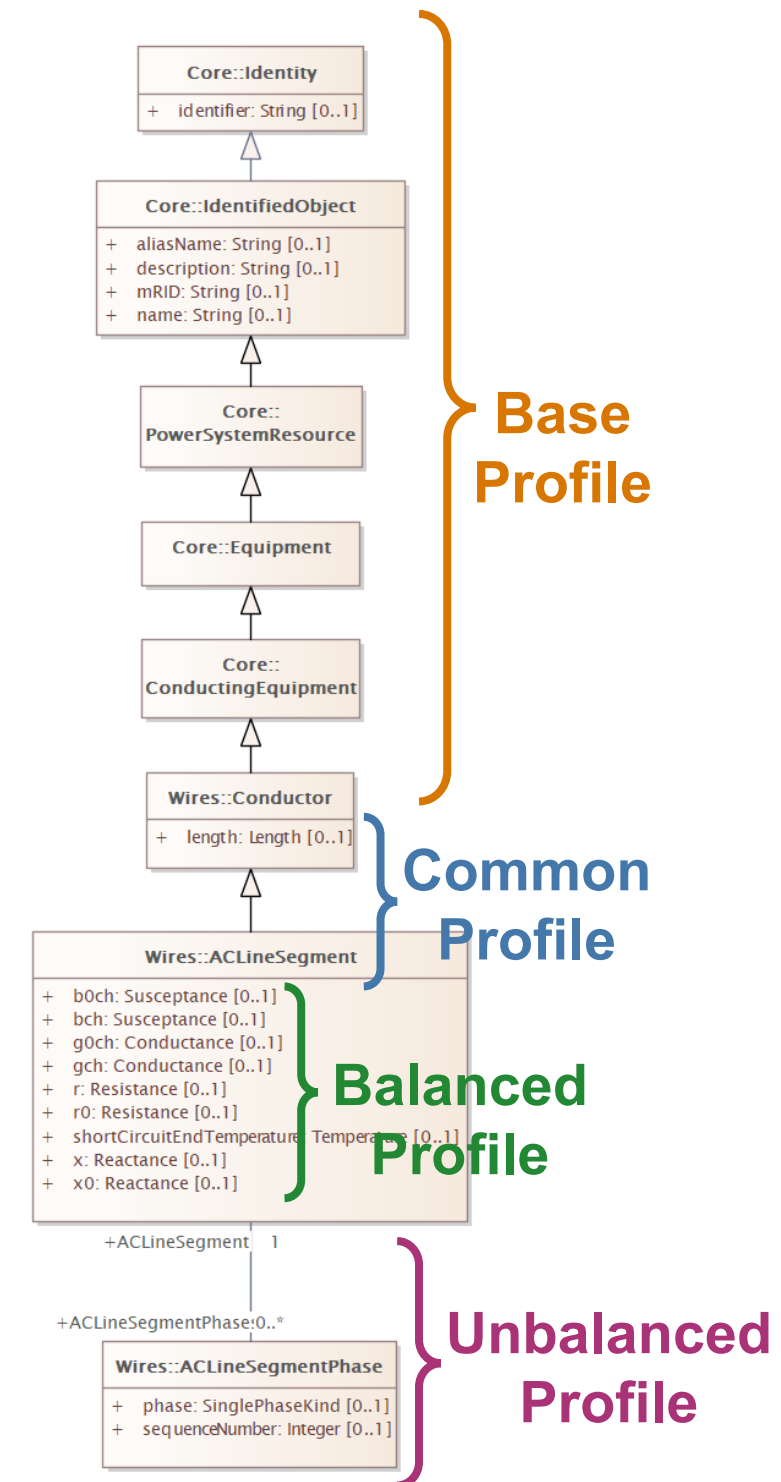
- CIM extends concepts of UML stereotypes for description of usage context
- Standard UML stereotypes:
 - **Class** – any “thing” that needs to be modeled
 - **enumeration** – list of things with fixed values (SinglePhaseKind allowed to be A, B, C)
- CIM Stereotypes
 - **Abstract Class** – parent class (ConductingEquipment) that will never appear in data
 - **Concrete Class** – data class (ACLineSegment) that will appear in instance data
 - **Primitive Class** – basic types aligned with XSD standard (Float, String, Boolean)
 - **CIMDatatype Class** – value classes with unit and multiplier (Voltage, Hours, Money)
 - **ShadowExtension** – new class with same name as canonical class for adding attributes
- Custom Stereotypes
 - Users can define custom stereotypes for naming of extensions with custom namespace (e.g. NC = European Network Codes)

CIM Stereotypes in Enterprise Architect



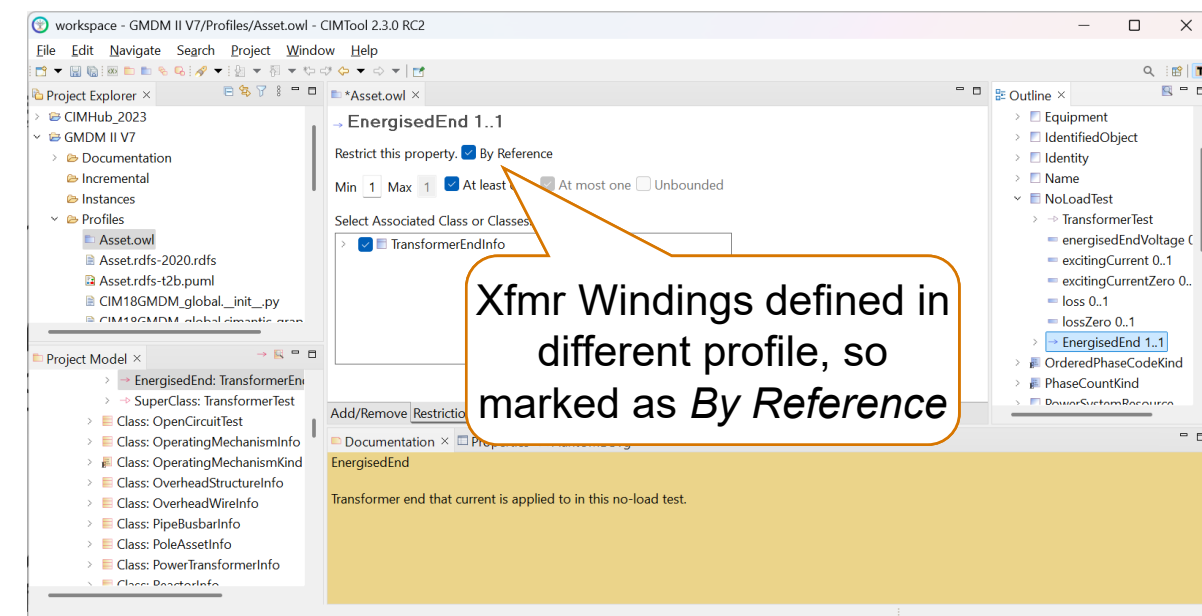
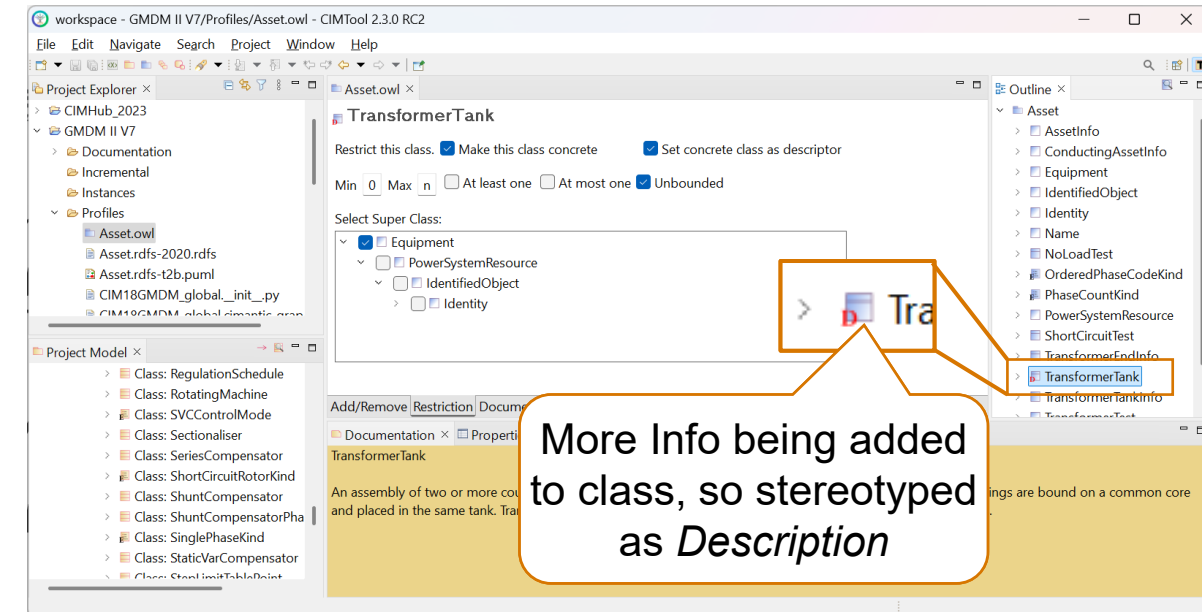
Next-Generation Profile Types

- Refined definitions of profiles to reduce ambiguity:
 - Base Profile** – Foundational collection of abstract classes needed for correct inheritance trees to be created
 - Variable Fidelity Modeling Block** – Snippets of class attributes / associations to be added to a Base Profile
 - Common Profile** – Standardized additive assembly of VFM Blocks for data exchange used by both TSO and DSO
 - Balanced Profile** – Assembly of VFM Blocks for TSO
 - Unbalanced Profile** – Assembly of VFM Blocks for DSO
 - Exchange Profile** – Custom assembly of VFM Blocks for clearly defined data exchange between two entities using a combination of common, balanced, and unbalanced profiles
 - Standard Profile** – Pre-defined exchange profile that is published in IEC 61970 / 61968 / 62325 / 62746 document



Formative vs Additive Data Definitions

- Formalizing definition of formative vs additive data exchanges:
 - Formative Definition** – The first time that an object is defined within exchange context. This will include name, mRID, identifier, etc.
 - ✓ Indicated in tooling by stereotyping class as Concrete within profile .owl file
 - Additive Definition** – Additional information from a different VFM block
 - ✓ Indicated in tooling by stereotyping class as Description within profile .owl file
 - By Reference** – The data is pointing to a class from a different VFM block that is not included in the data exchange file



Modeling Workflow





VFM Block Mappings

- 1:1 mappings developed between CIM classes and VFM Blocks to be created
- Mermaid.js diagrams created in markdown text description files that can be tracked via GitHub with versioning and change history

variable-fidelity-models / class_diagrams / Wires / ACLineSegment.md

Files

- feature/3
- SHACL
- cimvm
- class_diagrams
 - Core
 - Generation
 - Wires
 - ACLineSegment.md
 - ACLineSegmentPhase.md
 - AsynchronousMachine.md
 - EnergyConsumer.md
 - EnergyConsumerPhase.md
 - LinearShuntCompensator.md
 - LinearShuntCompensatorPhas...
 - PowerElectronicsConnection.md
 - PowerElectronicsConnectionPh...
 - PowerTransformer.md
 - PowerTransformerEnd.md
 - SynchronousMachine.md
 - TransformerEnd.md
 - TransformerTank.md
 - TransformerTankEnd.md
 - class_er_creator.ipynb
 - debug
 - dist
 - owl_editor
 - profile_diagrams
 - shacl_profile_builder
 - tests
 - .gitignore
 - LICENSE
 - README.md
 - poetry.lock
 - pyproject.toml

Usage

Type: Concrete

Formative: Connectivity>>Common

Appears in:

- Connectivity
 - Balanced
 - Unbalanced
- Electrical

Entity Relationship Diagrams

All attributes

Conductor		
float	length	Connectivity>>Line

Inherits from

ACLineSegment		
float	b0ch	Short Circuit>>Line
float	bch	Electrical>>Balanced>>Line
float	g0ch	Short Circuit>>Line
float	gch	Electrical>>Balanced>>Line
float	r	Electrical>>Balanced>>Line
float	r0	Short Circuit>>Line
float	shortCircuitEndTemperature	Short Circuit>>Line
float	x	Electrical>>Balanced>>Line
float	x0	Short Circuit>>Line
uuid	ACLineSegmentPhases	FK Connectivity>>Unbalanced
uuid	ParallelLineSegment	FK Connectivity>>TBD
uuid	PerLengthImpedance	FK Electrical>>Unbalance
uuid	WireSpacingInfo	FK Asset Association>>Line

Canonical Packages

Formative vs Additive Definitions

Parent Class

Attribute / Association

Datatype

Class

Blaming variable-fidelity-models

github.com/PNNL-CIM-Tools/variable-fidelity-models

variable-fidelity-models/class-diagrams/wires/ACLineSegment.mermaid

AAndersn vfblocks 26e57c1 · 3 weeks ago

63 lines (48 loc) · 1.39 KB

Preview Code Blame

Older Newer

Contributors 1

3 weeks ago update diagrams 1 #

2 months ago connectivity base 2

3 weeks ago update diagrams 3 ## Definition

4

5

6

7 ## Usage

8

9 Type: Concrete

10

11 Formative: Connectivity>>Common

12

3 weeks ago vfblocks 13 Appears in:

14 * Connectivity

15 * Balanced

16 * Unbalanced

17 * Electrical

18

3 weeks ago update diagrams 19

20 ## Entity Relationship Diagrams

21

22 ### All attributes

23

2 months ago connectivity base 24

25 ```mermaid

26 %%{init: {

27 "theme": "neutral",

28

29

30

31

32

33

34

35

36

37

38

39

40 ALineSegment{

41 float b0ch "Electrical>>Balanced>>Line"

42 float g0ch "Short Circuit>>Line"

43 float gch "Electrical>>Balanced>>Line"

44 float r "Electrical>>Balanced>>Line"

45 float r0 "Short Circuit>>Line"

46 float shortCircuitEndTemperature "Short

47 float x "Electrical>>Balanced>>Line"

48 float x0 "Short Circuit>>Line"

49 uuid ALineSegmentPhases FK "Connectivi

50 uuid ParallelLineSegment FK "Connectivi

51 uuid PerLengthImpedance FK "Electrical>

52 uuid WireSpacingInfo FK "Asset Associat

53 }

54 ALineSegment ||..|| Conductor : "Inherit

55 %% ALineSegment |o--o| ALineSegmentPhase

56 %% ALineSegment |o--o| ParallelLineSegment

57 %% ALineSegment |o--o| PerLengthImpedance

58 %% ALineSegment |o--o| WireSpacingInfo :

59

60 Conductor{

61 float length "Connectivity>>Line"

62 }

63

last month connectivity profiles

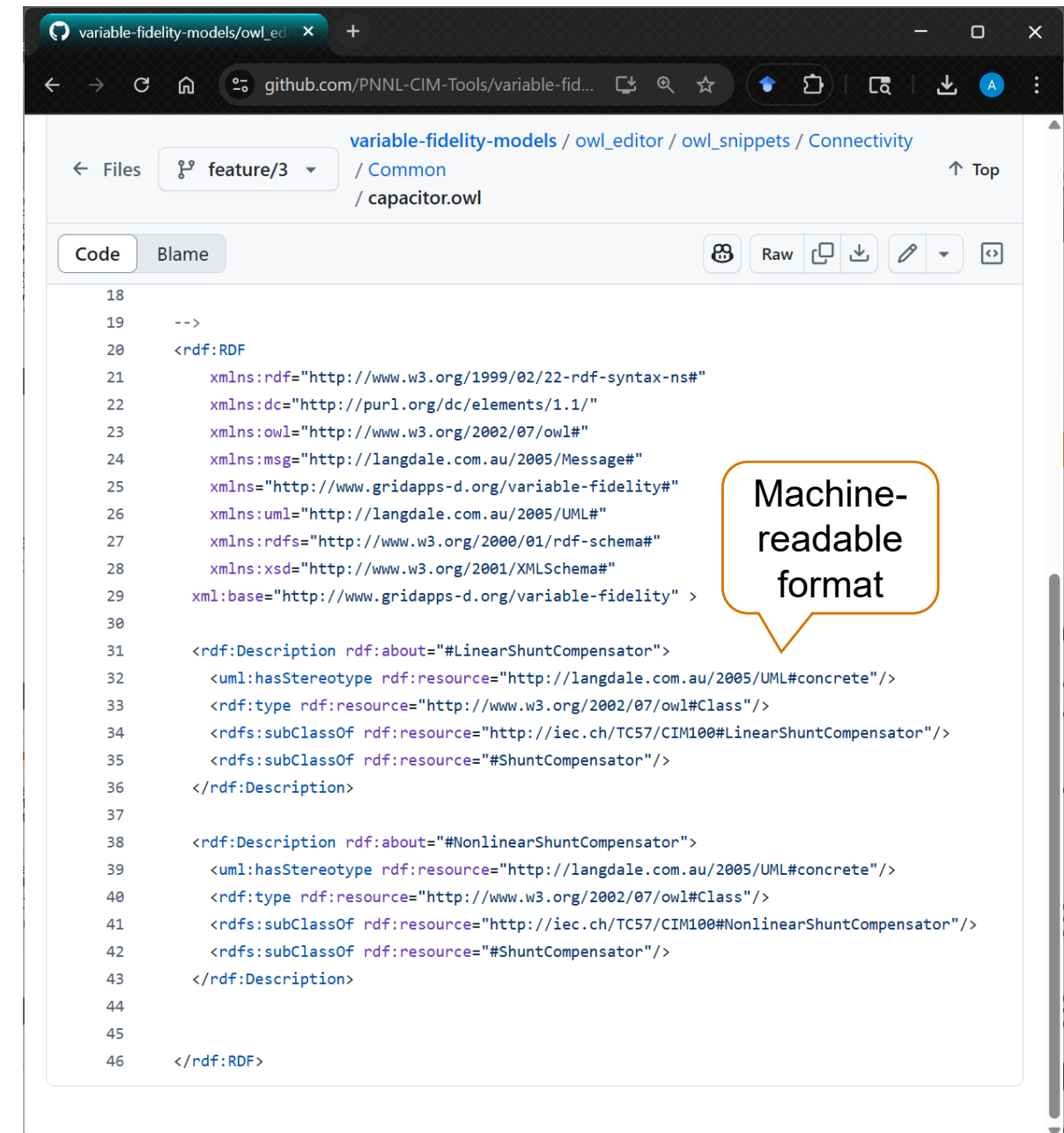
Machine-readable format

1 Mapping to VFM Block

Line-by-Line History via GitHub

VFM Block Definitions

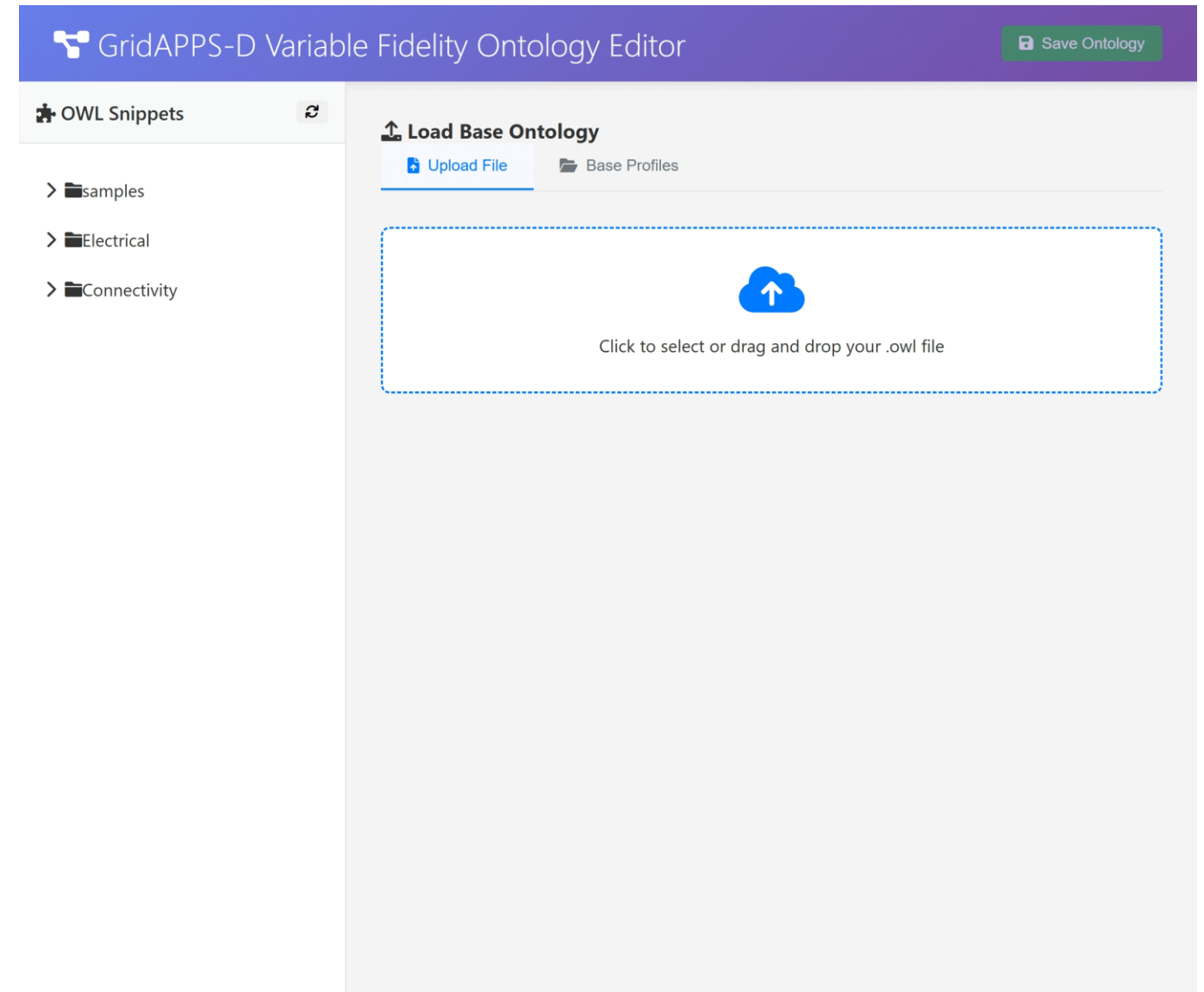
- VFM Blocks defined to align with new CIM18 distribution profiles
- Generated using CIMTool as .owl files, which are machine-readable artifacts that can be stacked additively to form custom exchange profiles
 - Connectivity
 - ✓ Common
 - Node, Terminal, Feeder, Substation, etc.
 - ✓ Balanced, Unbalanced
 - Line, Transformer, Capacitor, etc.
 - Electrical
 - ✓ Common, Balanced, Unbalanced
 - Line, Transformer, Capacitor, etc.
 - Asset Catalogue
 - Short Circuit
 - Dynamics



```
18
19 -->
20 <rdf:RDF
21   xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
22   xmlns:dc="http://purl.org/dc/elements/1.1/"
23   xmlns:owl="http://www.w3.org/2002/07/owl#"
24   xmlns:msg="http://langdale.com.au/2005/Message#"
25   xmlns="http://www.gridapps-d.org/variable-fidelity#"
26   xmlns:uml="http://langdale.com.au/2005/UML#"
27   xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
28   xmlns:xsd="http://www.w3.org/2001/XMLSchema#"
29   xml:base="http://www.gridapps-d.org/variable-fidelity" >
30
31   <rdf:Description rdf:about="#LinearShuntCompensator">
32     <uml:hasStereotype rdf:resource="http://langdale.com.au/2005/UML#concrete"/>
33     <rdf:type rdf:resource="http://www.w3.org/2002/07/owl#Class"/>
34     <rdfs:subClassOf rdf:resource="http://iec.ch/TC57/CIM100#LinearShuntCompensator"/>
35     <rdfs:subClassOf rdf:resource="#ShuntCompensator"/>
36   </rdf:Description>
37
38   <rdf:Description rdf:about="#NonlinearShuntCompensator">
39     <uml:hasStereotype rdf:resource="http://langdale.com.au/2005/UML#concrete"/>
40     <rdf:type rdf:resource="http://www.w3.org/2002/07/owl#Class"/>
41     <rdfs:subClassOf rdf:resource="http://iec.ch/TC57/CIM100#NonlinearShuntCompensator"/>
42     <rdfs:subClassOf rdf:resource="#ShuntCompensator"/>
43   </rdf:Description>
44
45
46 </rdf:RDF>
```

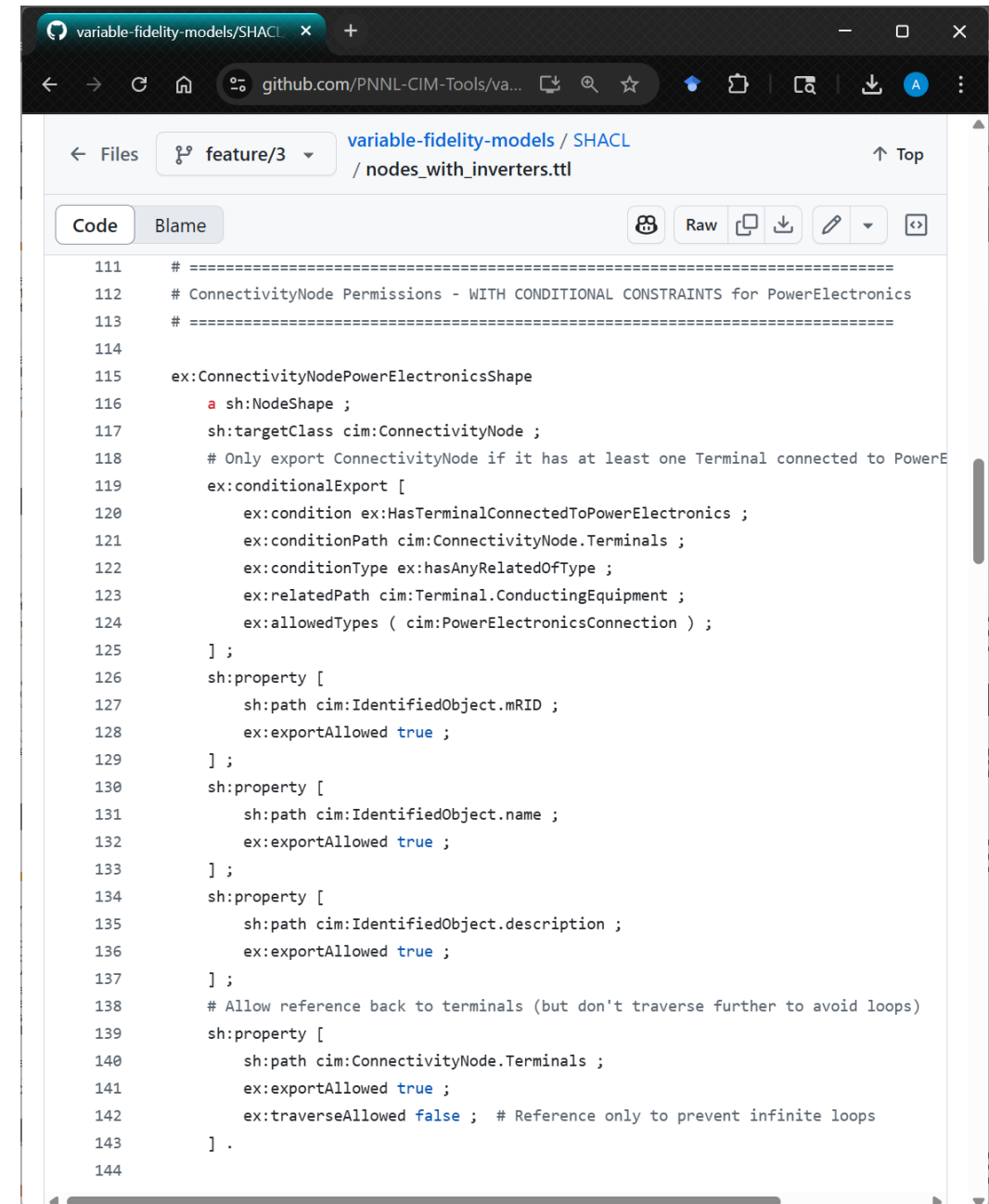
Variable Fidelity Ontology Editor

- GUI-based VFM Block editor developed for profile creation
- Drag-and-drop assembly of exchange profiles from pre-defined VFM Blocks
- Direct integration with CIMTool so that database schemas are auto-generated as soon as user saves merged ontology .owl file into workspace
- Will be packaged and released as desktop app after final cleanup of CIM17 and CIM18 VFM blocks



SHACL Constraints on Data Values

- Shapes Constraint Language (SHACL) is standards-based schema for specifying constraints on classes and property paths
- Increasingly being used for data validation by specifying rules for values, cardinality, etc. for ENTSO-E CIM exchange profiles
- Developed extension framework allowing plain-text specification of restrictions of data that can be exported
- Example at right specifies that node can only be exported if it is connected to an inverter



```
111 # =====
112 # ConnectivityNode Permissions - WITH CONDITIONAL CONSTRAINTS for PowerElectronics
113 # =====
114
115 ex:ConnectivityNodePowerElectronicsShape
116   sh:NodeShape ;
117   sh:targetClass cim:ConnectivityNode ;
118   # Only export ConnectivityNode if it has at least one Terminal connected to PowerE
119   ex:conditionalExport [
120     ex:condition ex:HasTerminalConnectedToPowerElectronics ;
121     ex:conditionPath cim:ConnectivityNode.Terminals ;
122     ex:conditionType ex:hasAnyRelatedOfType ;
123     ex:relatedPath cim:Terminal.ConductingEquipment ;
124     ex:allowedTypes ( cim:PowerElectronicsConnection ) ;
125   ] ;
126   sh:property [
127     sh:path cim:IdentifiedObject.mRID ;
128     ex:exportAllowed true ;
129   ] ;
130   sh:property [
131     sh:path cim:IdentifiedObject.name ;
132     ex:exportAllowed true ;
133   ] ;
134   sh:property [
135     sh:path cim:IdentifiedObject.description ;
136     ex:exportAllowed true ;
137   ] ;
138   # Allow reference back to terminals (but don't traverse further to avoid loops)
139   sh:property [
140     sh:path cim:ConnectivityNode.Terminals ;
141     ex:exportAllowed true ;
142     ex:traverseAllowed false ; # Reference only to prevent infinite loops
143   ] .
144
```

CIMantic Graphs SHACL-based Export

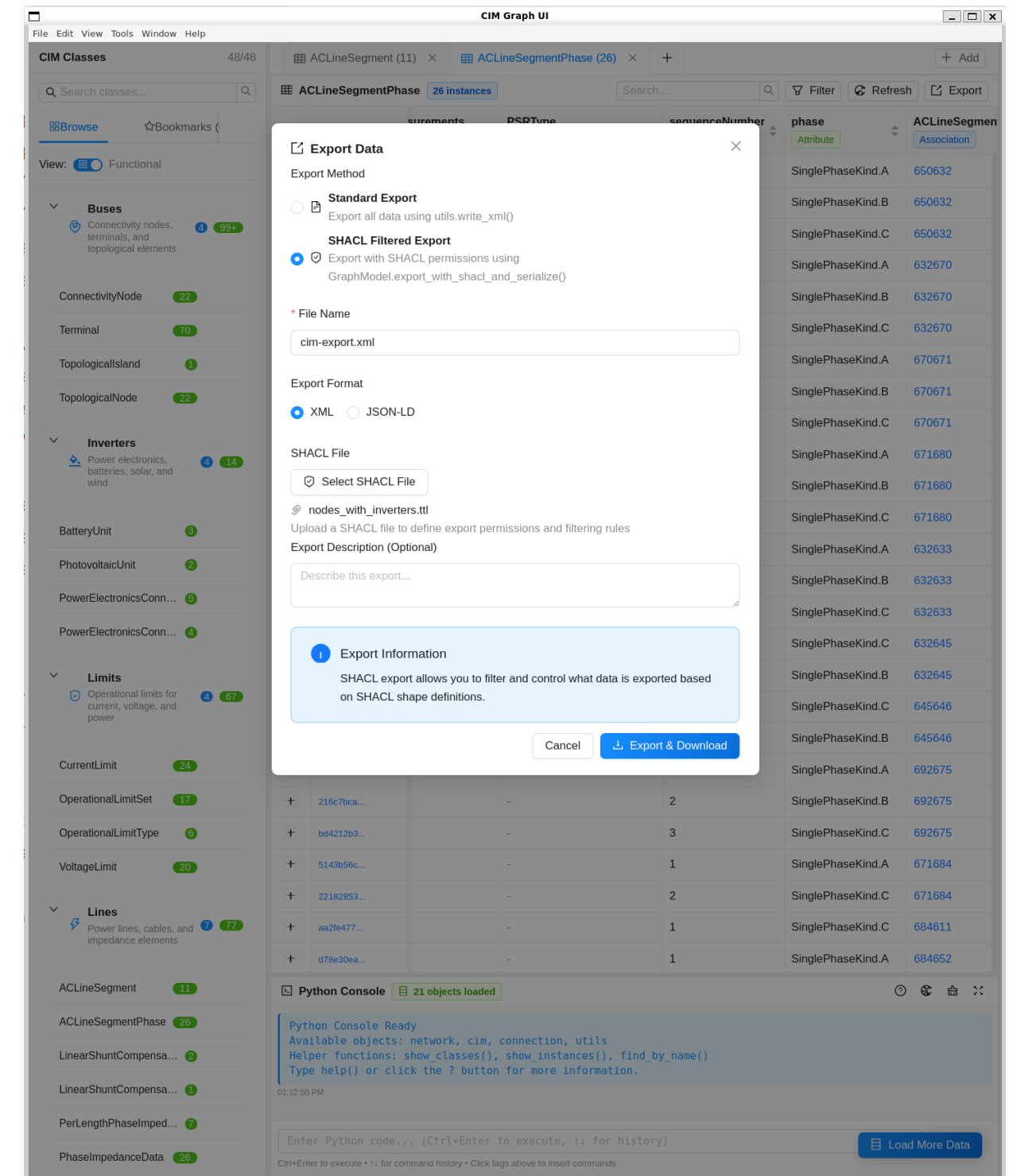
- CIMantic Graphs 0.4.0 release now supports SHACL-based control of variable fidelity data exports
- Simple two step-process of opening feeder model and then saving with SHACL rules
- XML and JSON-LD output supported

```
1 from cimgraph.databases import XMLFile
2 from cimgraph.models import FeederModel
✓ 0.6s Python

1 file = XMLFile(filename='../tests/test_models/final9500node_CIM100x.XML')
2 network = FeederModel(container=cim.Feeder(), connection=file)
✓ 14.6s Python

1 network.export_with_shacl_and_serialize('../SHACL/nodes_with_inverters.ttl',
2                                         'filtered_output.xml', format='xml')
✓ 0.6s Python
```

CIM-Graph Python API Export (5 lines of code)



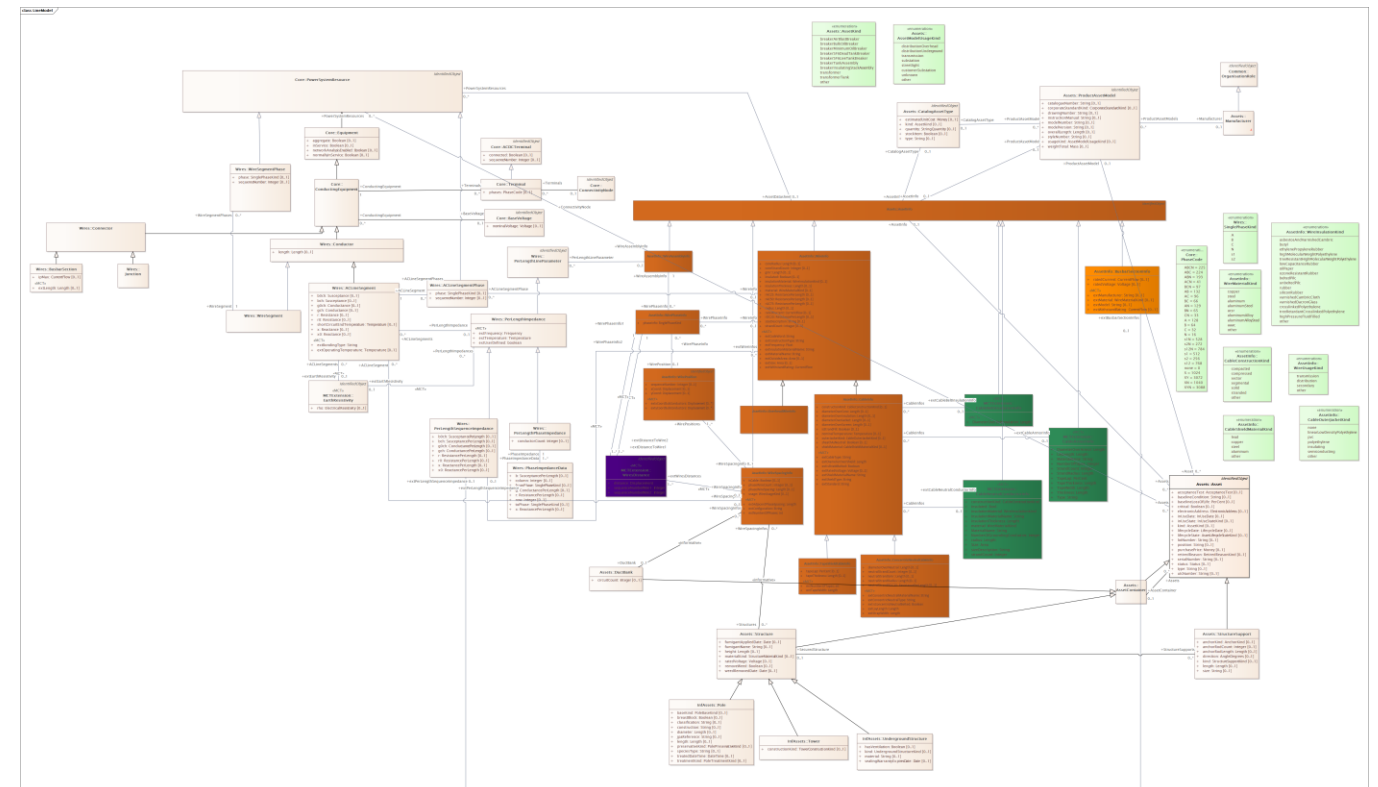
CIMantic Studio Desktop App Export (new)

Next Steps



Missing Classes in CIM

- Significant gaps were identified in CIM classes for unbalanced feeders with underground cables, overhead distribution lines, load modeling
- Launching new IEC / UCAiug WG13 task force to identify / add all classes needed for effective model conversion between power flow solvers
- Collaborating with EPRI, VELCO, ERCOT, Hitachi, AspenTech, etc. to create comprehensive model
- New representations will be first used in EPRI MCT / CIMHub and Atlas tooling as CIM extensions and then proposed for “golden copy” CIM 18



Missing cable classes identified by EPRI team

Defining Data Exchange Metadata Concepts

- Simple Knowledge Organization System (SKOS) is W3C standard that could be used to define use case specific exchange profiles
- Could use along with DublinCore and Data Catalog Vocabulary (DCAT) standards to define complete metadata regarding
 - Model versioning and data provenance
 - High / medium / low fidelity modeling concepts
 - Formative vs additive object definitions
 - Data source (measured vs calculated)
 - Criticality of data (essential / important / optional)
 - Allowable consumers of data
 - Etc.

```
Code Blame
1  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
2  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
3  @prefix skos: <http://www.w3.org/2004/02/skos/core#> .
4  @prefix dcterms: <http://purl.org/dc/terms/> .
5  @prefix vfm: <http://gridappsd.org/variable-fidelity#> .
6
7  # VFM Concept Scheme
8  <http://gridappsd.org/variable-fidelity/concepts> a skos:ConceptScheme ;
9    dcterms:title "Variable Fidelity Modeling Concepts" ;
10   dcterms:description "Conceptual classification for variable fidelity grid model data" ;
11   dcterms:creator "UCAiUG CIM Users Group" ;
12   vfm:supportsFidelityLevels vfm:HighFidelity, vfm:MediumFidelity, vfm:LowFidelity .
13
14 # Model Type Concepts
15 vfm:ConnectivityModel a skos:Concept ;
16   skos:inScheme <http://gridappsd.org/variable-fidelity/concepts> ;
17   skos:prefLabel "Connectivity Model" ;
18   skos:definition "Model data describing equipment connectivity and topology" ;
19   skos:topConceptOf <http://gridappsd.org/variable-fidelity/concepts> ;
20   vfm:requiredForFidelity vfm:LowFidelity, vfm:MediumFidelity, vfm:HighFidelity .
21
22 vfm:GeographicalModel a skos:Concept ;
23   skos:inScheme <http://gridappsd.org/variable-fidelity/concepts> ;
24   skos:prefLabel "Geographical Model" ;
25   skos:definition "Model data describing geographical location and spatial relationships" ;
26   skos:topConceptOf <http://gridappsd.org/variable-fidelity/concepts> ;
27   vfm:requiredForFidelity vfm:MediumFidelity, vfm:HighFidelity .
28
29 vfm:ElectricalModel a skos:Concept ;
30   skos:inScheme <http://gridappsd.org/variable-fidelity/concepts> ;
31   skos:prefLabel "Electrical Model" ;
32   skos:definition "Model data describing electrical characteristics and parameters" ;
33   skos:topConceptOf <http://gridappsd.org/variable-fidelity/concepts> ;
34   vfm:requiredForFidelity vfm:HighFidelity .
35
36 # Export Control Concepts
37 vfm:ExportControlModel a skos:Concept ;
38   skos:inScheme <http://gridappsd.org/variable-fidelity/concepts> ;
39   skos:prefLabel "Export Control Model" ;
40   skos:definition "Rules and policies governing data export based on fidelity requirements" ;
41   skos:topConceptOf <http://gridappsd.org/variable-fidelity/concepts> .
42
43 # Fidelity-specific subconcepts
44 vfm:HighFidelityTopology a skos:Concept ;
45   skos:inScheme <http://gridappsd.org/variable-fidelity/concepts> ;
46   skos:prefLabel "High Fidelity Topology" ;
47   skos:broader vfm:ConnectivityModel ;
48   skos:definition "Detailed connectivity model with all switching devices and connection poin
49
50 vfm:SimplifiedTopology a skos:Concept ;
51   skos:inScheme <http://gridappsd.org/variable-fidelity/concepts> ;
52   skos:prefLabel "Simplified Topology" ;
53   skos:broader vfm:ConnectivityModel ;
54   skos:definition "Simplified connectivity model focusing on major equipment only" .
```

Thank you



DublinCore + SKOS Implementation Concept

Formative Line Object

```
Code Blame
<?xml version="1.0" encoding="UTF-8"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
  xmlns:dcterms="http://purl.org/dc/terms/"
  xmlns:cim="http://iec.ch/TC57/CIM100#"
  xmlns:vfm="http://gridappsd.org/variable-fidelity#"
  xmlns:foaf="http://xmlns.com/foaf/0.1/">

  <!-- Mini Version -->
  <cim:ACLineSegment rdf:about="urn:uuid:AAA40048-0566-45CD-A3F1-71D52B3E9BD1">
    <cim:ConductingEquipment.BaseVoltage rdf:resource="urn:uuid:A1A90D64-224D-4E25-8BEE-FA7A518CA4A3"/>
    <cim:Equipment.EquipmentContainer rdf:resource="urn:uuid:F7AA0E50-429C-4F62-8786-888AD96A42B4"/>
    <cim:Identity.Identifier>AAA40048-0566-45CD-A3F1-71D52B3E9BD1</cim:Identity.Identifier>
    <!-- Definition Metadata -->
    <dcterms:type rdf:resource="http://gridappsd.org/variable-fidelity#FormativeDefinition"/>
    <dcterms:conformsTo rdf:resource="http://cim.ucaiug.io/gmdm/profiles/connectivity/unbalanced"/>
  </cim:ACLineSegment>

  <!-- Full Version with VFM Metadata -->
  <cim:ACLineSegment rdf:about="urn:uuid:000DBDC9-F896-4332-B4A0-0CB54C939C42">
    <!-- Core connectivity properties -->
    <cim:ConductingEquipment.BaseVoltage rdf:resource="urn:uuid:A1A90D64-224D-4E25-8BEE-FA7A518CA4A3"/>
    <cim:Equipment.EquipmentContainer rdf:resource="urn:uuid:F7AA0E50-429C-4F62-8786-888AD96A42B4"/>
    <cim:IdentifiedObject.mRID>000DBDC9-F896-4332-B4A0-0CB54C939C42</cim:IdentifiedObject.mRID>
    <cim:Identity.Identifier>000DBDC9-F896-4332-B4A0-0CB54C939C42</cim:Identity.Identifier>

    <!-- VFM Definition Metadata -->
    <dcterms:type rdf:resource="http://gridappsd.org/variable-fidelity#FormativeDefinition"/>
    <dcterms:conformsTo rdf:resource="http://cim.ucaiug.io/gmdm/profiles/connectivity/unbalanced"/>
    <dcterms:subject rdf:resource="http://cim.ucaiug.io/vfm/concepts/ConnectivityModel"/>
    <dcterms:created rdf:datatype="http://www.w3.org/2001/XMLSchema#dateTime">2024-01-15T10:30:00Z</dcterms:created>
    <dcterms:creator>
      <!-- Replace with cim:ModelAuthority -->
      <foaf:Agent>
        <foaf:name>Grid Model Management System</foaf:name>
        <vfm:systemComponent>Connectivity Processor</vfm:systemComponent>
      </foaf:Agent>
    </dcterms:creator>
    <dcterms:source>Network Topology Database</dcterms:source>
    <dcterms:isRequiredBy rdf:resource="http://cim.ucaiug.io/vfm/requirements/PowerFlowAnalysis"/>
    <vfm:definitionStatus rdf:resource="http://gridappsd.org/variable-fidelity#Complete"/>
    <vfm:dataQuality rdf:resource="http://gridappsd.org/variable-fidelity#Validated"/>
    <vfm:criticality rdf:resource="http://gridappsd.org/variable-fidelity#Essential"/>
    <vfm:fidelityLevel rdf:resource="http://gridappsd.org/variable-fidelity#HighFidelity"/>
  </cim:ACLineSegment>
```

Additive Line Object

```
Code Blame
<!-- Mini Version -->
<cim:ACLineSegment rdf:about="urn:uuid:AAA40048-0566-45CD-A3F1-71D52B3E9BD1">
  <cim:ACLineSegment.PerLengthImpedance rdf:resource="urn:uuid:B4FF5879-ABE6-4CF2-92D7-38987F9269B6"/>
  <cim:Conductor.length>42.081798</cim:Conductor.length>
  <!-- Definition Metadata -->
  <dcterms:type rdf:resource="http://gridappsd.org/variable-fidelity#AdditiveDefinition"/>
  <dcterms:conformsTo rdf:resource="http://cim.ucaiug.io/gmdm/profiles/electrical/unbalanced"/>
</cim:ACLineSegment>

<!-- Full version with complete metadata -->
<cim:ACLineSegment rdf:about="urn:uuid:000DBDC9-F896-4332-B4A0-0CB54C939C42">
  <!-- Electrical properties -->
  <cim:ACLineSegment.PerLengthImpedance rdf:resource="urn:uuid:12994F42-5878-4DE6-AAB4-81245B1C0899"/>
  <cim:Conductor.length>148.62445</cim:Conductor.length>

  <!-- VFM Definition Metadata -->
  <dcterms:type rdf:resource="http://gridappsd.org/variable-fidelity#AdditiveDefinition"/>
  <dcterms:conformsTo rdf:resource="http://cim.ucaiug.io/gmdm/profiles/electrical/unbalanced"/>
  <dcterms:subject rdf:resource="http://cim.ucaiug.io/vfm/concepts/ElectricalModel"/>
  <dcterms:created rdf:datatype="http://www.w3.org/2001/XMLSchema#dateTime">2024-01-15T12:45:00Z</dcterms:created>
  <dcterms:creator>
    <foaf:Agent>
      <foaf:name>Engineering Analysis System</foaf:name>
      <vfm:systemComponent>Electrical Parameter Processor</vfm:systemComponent>
    </foaf:Agent>
  </dcterms:creator>
  <dcterms:source>Equipment Testing Database</dcterms:source>
  <dcterms:requires rdf:resource="urn:uuid:000DBDC9-F896-4332-B4A0-0CB54C939C42"/>
  <dcterms:relation>
    <!-- Metadata for extensions as relevant -->
    <vfm:ProfileRelation>
      <vfm:extendsProfile rdf:resource="http://cim.ucaiug.io/gmdm/profiles/connectivity/unbalanced">
      <vfm:relationshipType rdf:resource="http://cim.ucaiug.io/gmdm/vocab#ElectricalExtension"/>
    </vfm:ProfileRelation>
  </dcterms:relation>
  <vfm:definitionStatus rdf:resource="http://gridappsd.org/variable-fidelity#Complete"/>
  <vfm:dataQuality rdf:resource="http://gridappsd.org/variable-fidelity#Measured"/>
  <vfm:criticality rdf:resource="http://gridappsd.org/variable-fidelity#Critical"/>
  <vfm:fidelityLevel rdf:resource="http://gridappsd.org/variable-fidelity#HighFidelity"/>
  <vfm:lastValidated rdf:datatype="http://www.w3.org/2001/XMLSchema#dateTime">2024-01-10T09:00:00Z</vfm:lastValidated>
</cim:ACLineSegment>
```

Profile Definitions

```
Code Blame
1 @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
2 @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
3 @prefix dcterms: <http://purl.org/dc/terms/> .
4 @prefix dcat: <http://www.w3.org/ns/dcat#> .
5 @prefix vfm: <http://gridappsd.org/variable-fidelity/vocab#> .
6 @prefix sh: <http://www.w3.org/ns/shacl#> .
7
8 # Connectivity Profile with Export Controls
9 <http://cim.ucaiug.io/gmdm/profiles/connectivity/unbalanced> a dcat:Dataset, vfm:FormativeProfile ;
10 dcterms:title "VFM Connectivity Unbalanced Profile" ;
11 dcterms:description "Core connectivity model for unbalanced power system analysis with export controls" ;
12 dcterms:identifier "vfm-connectivity-unbalanced-v1.0" ;
13 dcat:keyword "connectivity", "topology", "unbalanced", "vfm" ;
14 dcterms:conformsTo <http://iec.ch/TC57/ns/CIM/prof-cim#> ;
15 vfm:profileType vfm:FormativeProfile ;
16 vfm:definitionStatus vfm:Complete ;
17 vfm:fidelityLevel vfm:HighFidelity ;
18 vfm:hasExportPolicy vfm:BreakerExportPolicy, vfm:TerminalExportPolicy, vfm:ConnectivityNodeExportPolicy ;
19 dcterms:created "2024-01-15"^^xsd:date .
20
21 # Geographical Location Profile
22 <http://cim.ucaiug.io/gmdm/profiles/geographical_location> a dcat:Dataset, vfm:AdditiveProfile ;
23 dcterms:title "VFM Geographical Location Profile" ;
24 dcterms:description "Geographic location and spatial information with variable fidelity support" ;
25 dcterms:identifier "vfm-geographical-location-v1.0" ;
26 dcat:keyword "geography", "location", "spatial", "vfm" ;
27 dcterms:conformsTo <http://iec.ch/TC57/ns/CIM/prof-cim#> ;
28 vfm:profileType vfm:AdditiveProfile ;
29 vfm:extendsProfile <http://cim.ucaiug.io/gmdm/profiles/connectivity/unbalanced> ;
30 vfm:fidelityLevel vfm:MediumFidelity ;
31 dcterms:created "2024-01-15"^^xsd:date .
32
33 # Electrical Unbalanced Profile with Enhanced Export Controls
34 <http://cim.ucaiug.io/gmdm/profiles/electrical/unbalanced> a dcat:Dataset, vfm:AdditiveProfile ;
35 dcterms:title "VFM Electrical Unbalanced Profile" ;
36 dcterms:description "Electrical parameters for unbalanced analysis with fidelity controls" ;
37 dcterms:identifier "vfm-electrical-unbalanced-v1.0" ;
38 dcat:keyword "electrical", "impedance", "unbalanced", "vfm" ;
39 dcterms:conformsTo <http://iec.ch/TC57/ns/CIM/prof-cim#> ;
40 vfm:profileType vfm:AdditiveProfile ;
41 vfm:extendsProfile <http://cim.ucaiug.io/gmdm/profiles/connectivity/unbalanced> ;
42 vfm:fidelityLevel vfm:HighFidelity ;
43 vfm:criticality vfm:Critical ;
44 dcterms:created "2024-01-15"^^xsd:date .
```